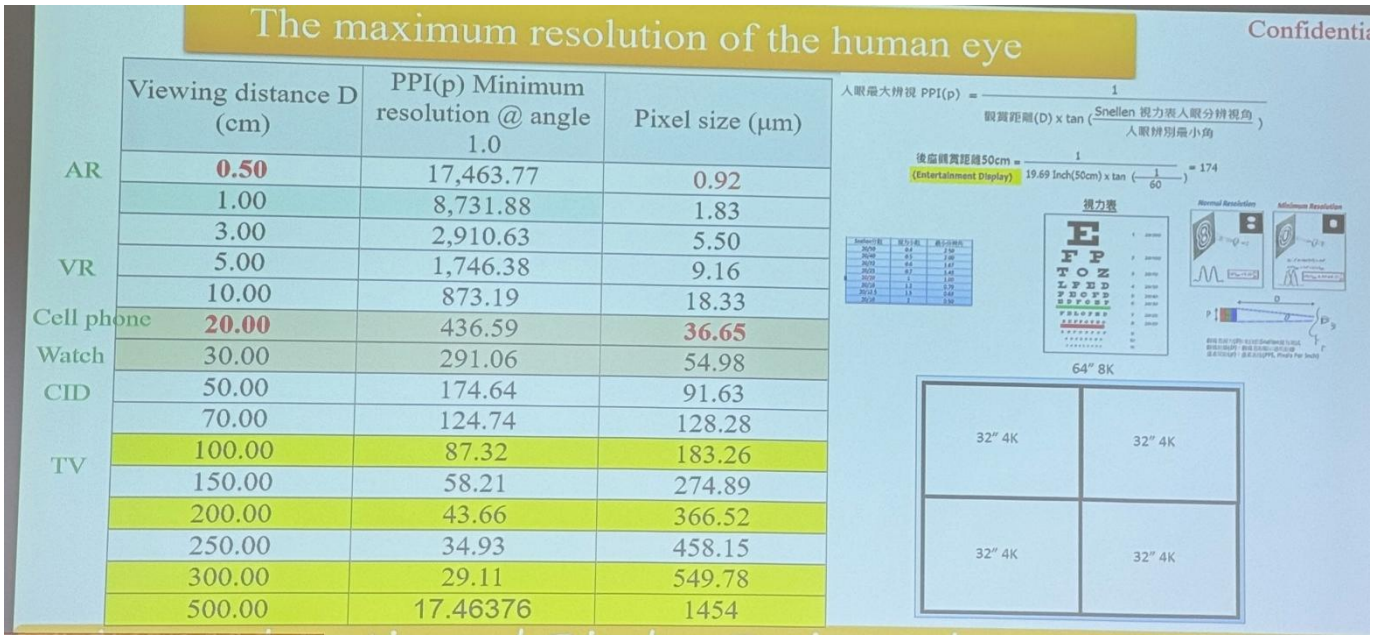
日期:2026/03/03　　講者:葉志庭

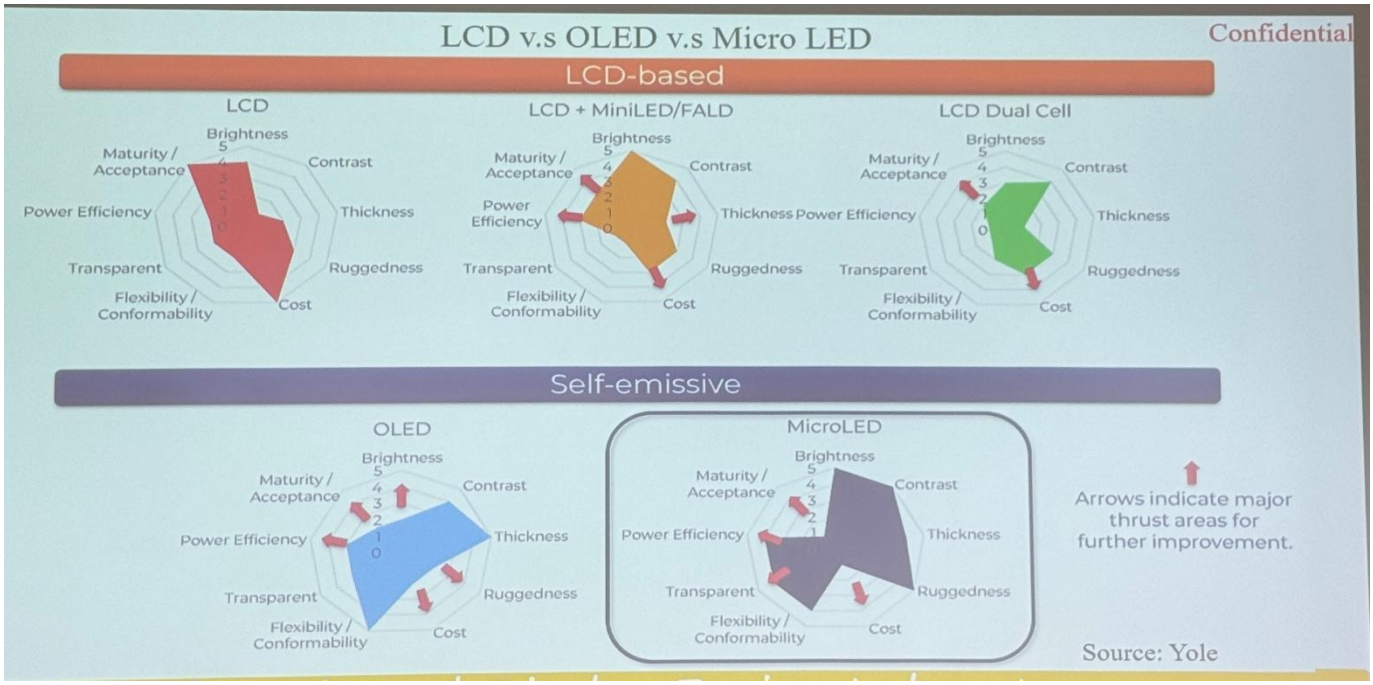
題目:Mini/Micro LED Design and Challenges for Advanced Displays



	Viewing distance D (cm)	PPI(p) Minimum resolution @ angle 1.0	Pixel size (μm)
AR	0.50	17,463.77	0.92
	1.00	8,731.88	1.83
	3.00	2,910.63	5.50
VR	5.00	1,746.38	9.16
	10.00	873.19	18.33
Cell phone	20.00	436.59	36.65
Watch	30.00	291.06	54.98
CID	50.00	174.64	91.63
	70.00	124.74	128.28
TV	100.00	87.32	183.26
	150.00	58.21	274.89
	200.00	43.66	366.52
	250.00	34.93	458.15
	300.00	29.11	549.78
	500.00	17.46376	1454

人眼極限解析度

- 越近 PPI 越高
- AR/VR（超近距）需要超高 PPI
- TV（遠距）需要 PPI 低



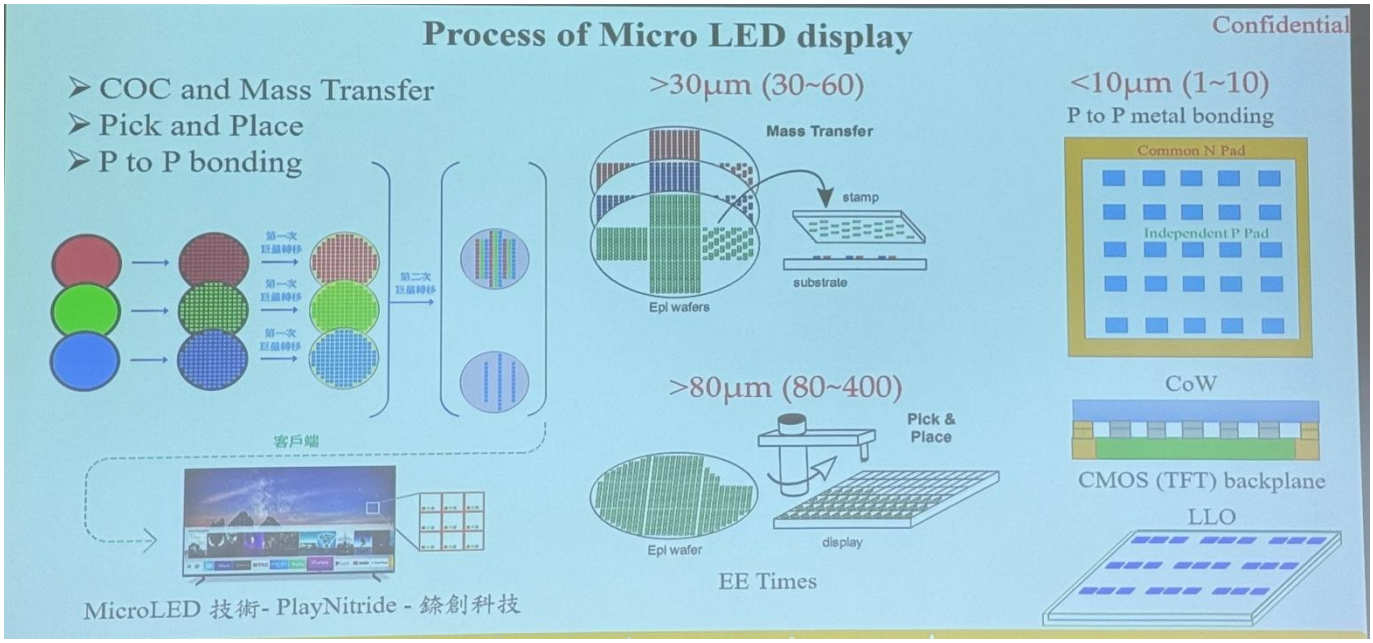
LCD-based

- LCD :便宜、成熟，但對比受限
- LCD + MiniLED/FALD: 亮度/對比提升
- Dual cell LCD: 使用兩層液晶去撐對比，厚度和成本代價高

Self-emissive

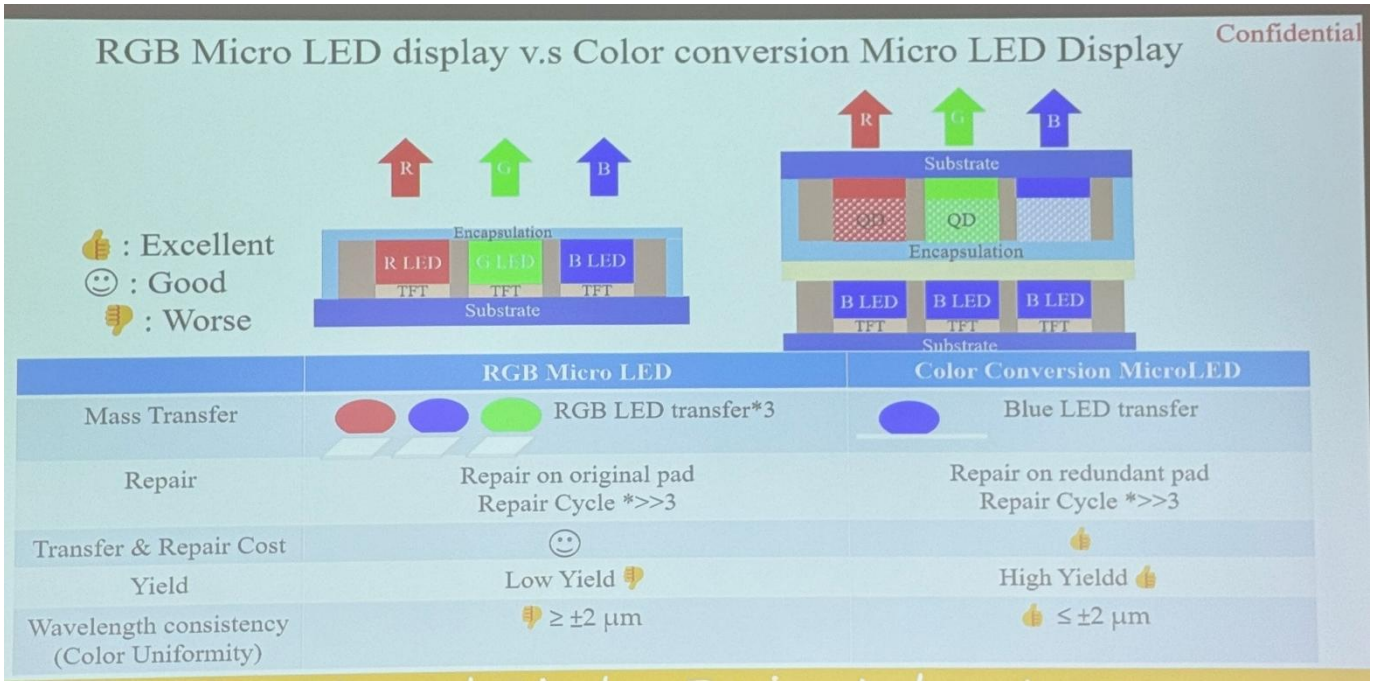
- OLED: 對比佳、薄，但亮度和壽命低

- Micro LED: 亮度、壽命、效率潛力好，但量產不易



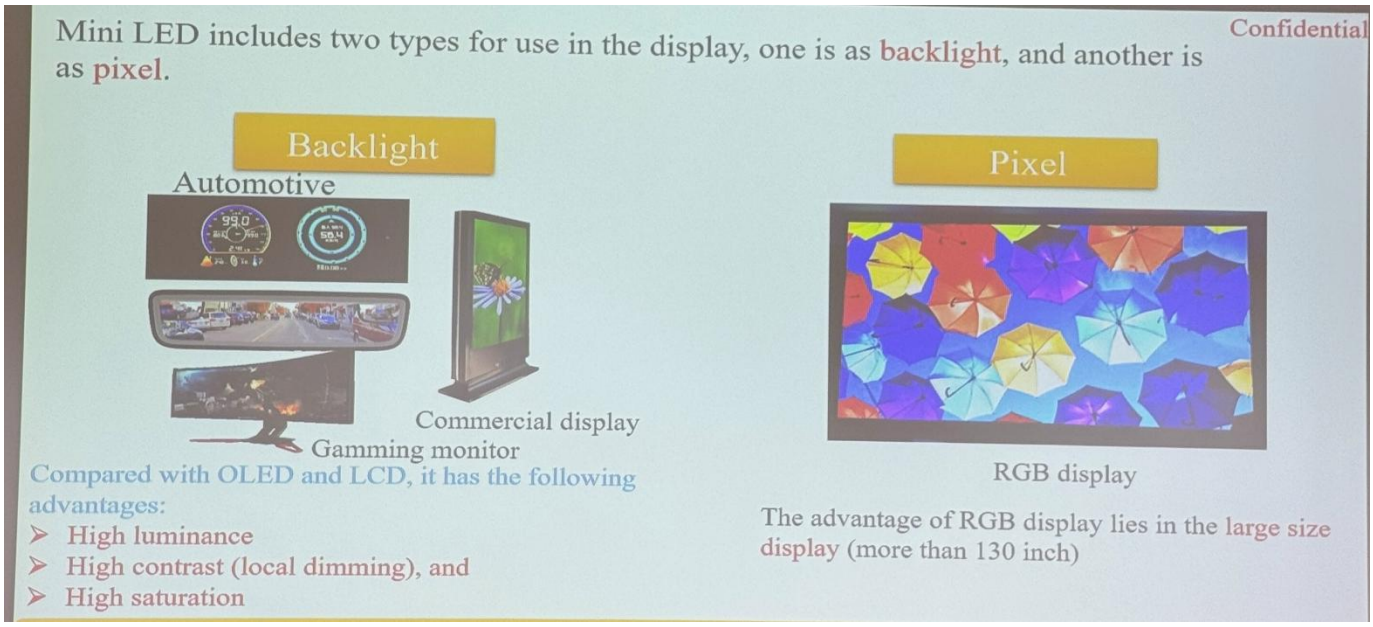
1. COC + Mass Transfer（晶粒大量轉移）
2. Pick & Place（放置/組裝）
3. P to P bonding（Pad-to-Pad 金屬鍵結）

- CoW (Chip on Wafer)：在晶圓上先做/先整理晶粒
- COC1 / COC2 (Chip on Carrier)：把晶粒先轉到載板上重排，再轉到最終背板/面板



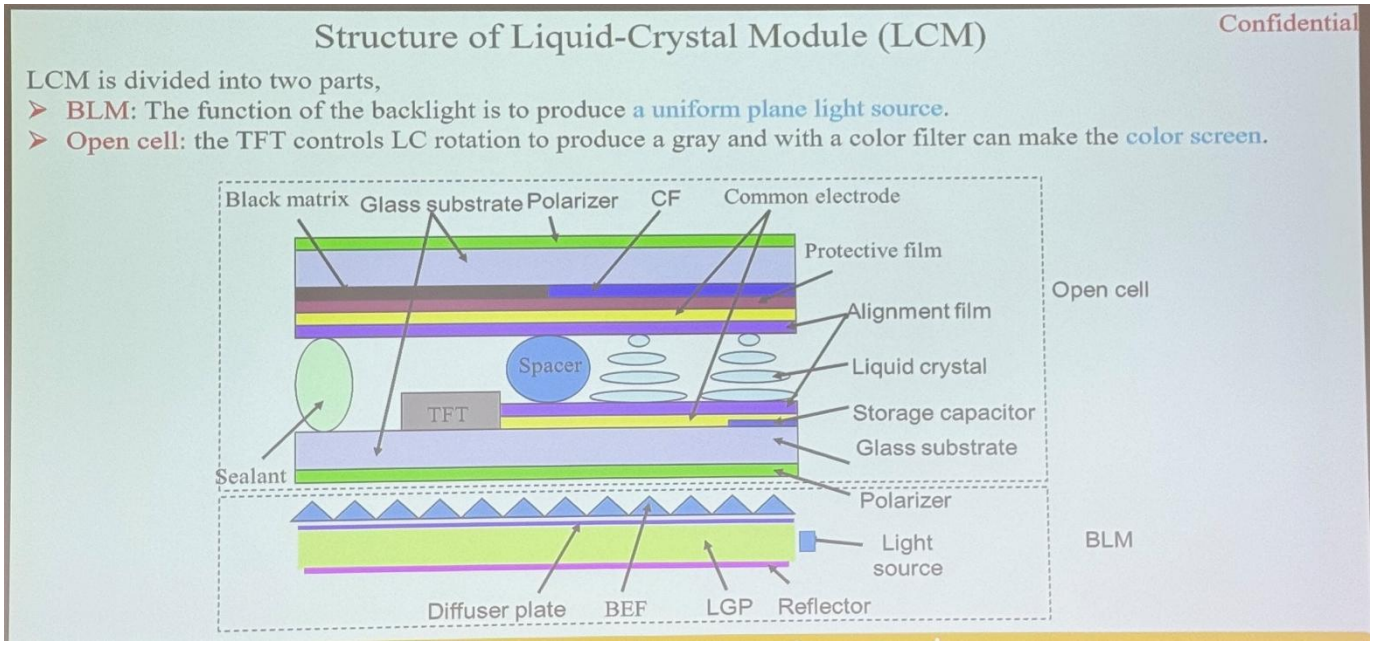
RGB MicroLED：R/G/B 三色都要轉移，成本高、良率低，色彩一致性差

Color Conversion: 只轉移 Blue LED，QD 螢光材料轉成 R/G，只要搬藍色，轉移次數少、良率較好、成本較低

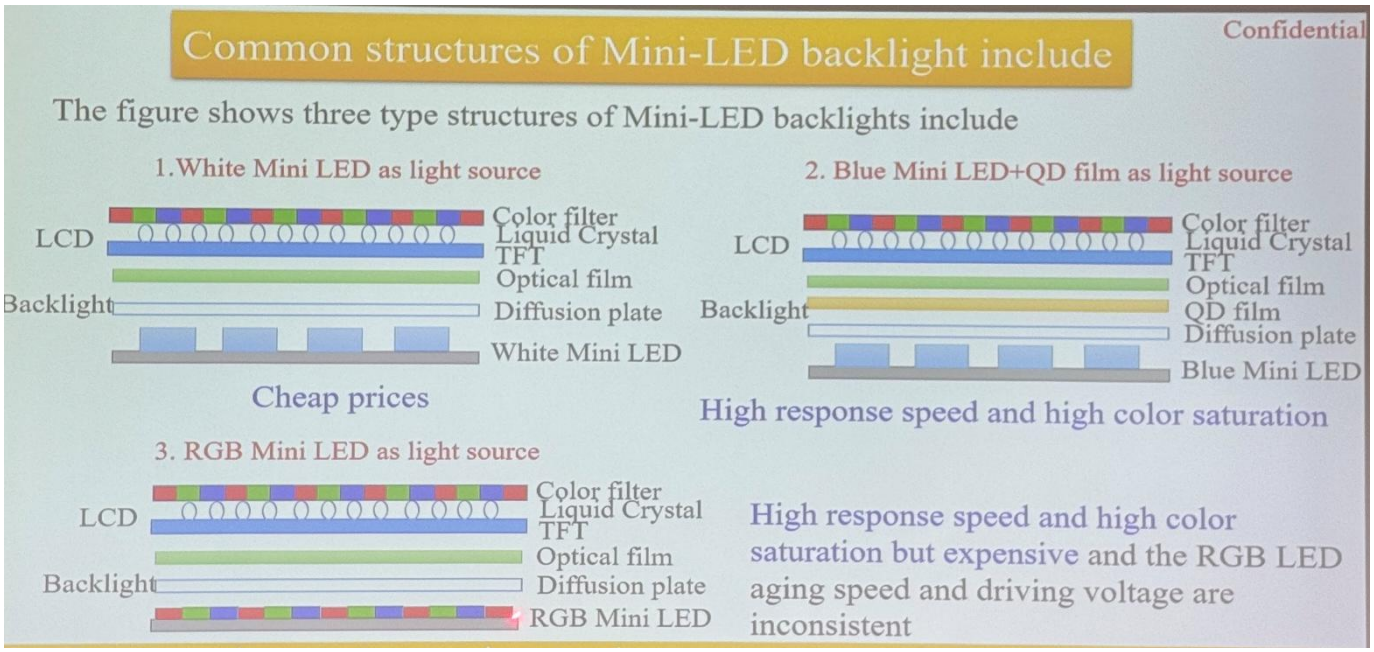


Mini LED

- 背光（Backlight）：車用、螢幕、電競顯示器
  - 高亮度、分區控光提升對比、飽和度
- 像素（Pixel / RGB display）：超大尺寸(>130 inch)

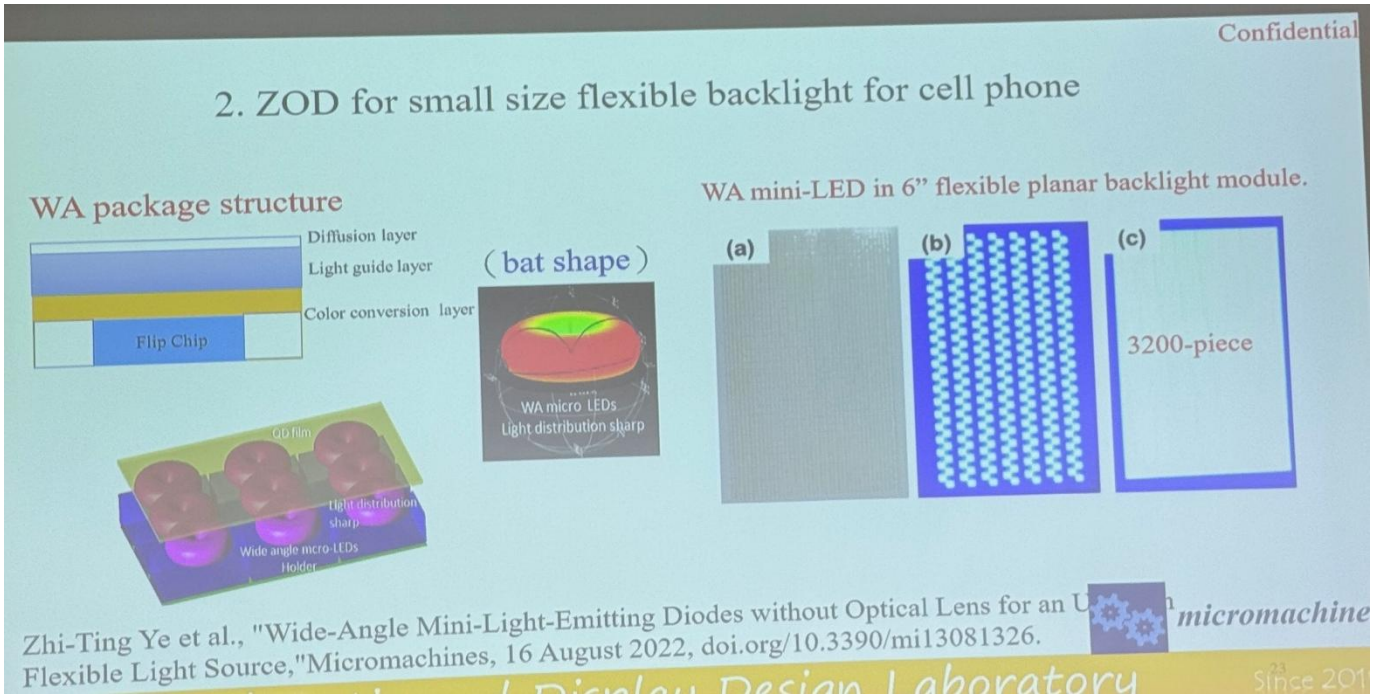


100%的光從 source 進去，最後只剩 4%出來



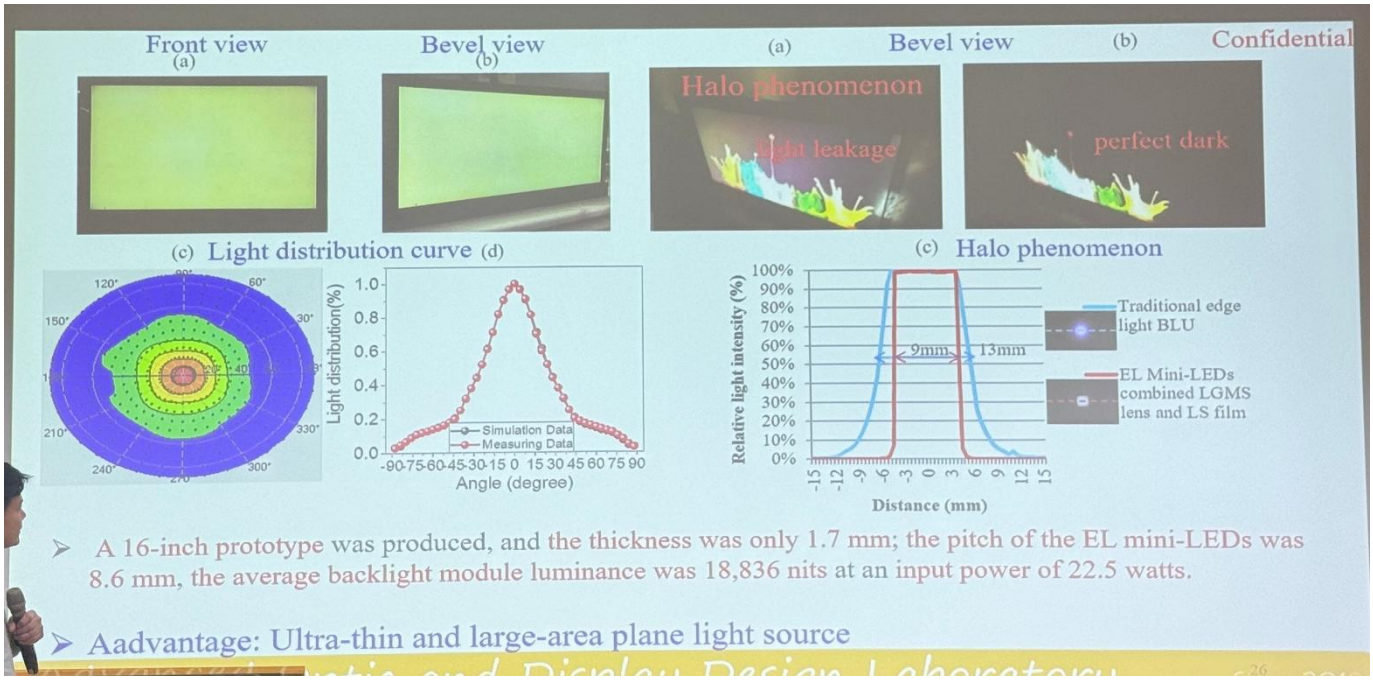
1. White Mini-LED
   - 結構簡單
   - 成本最低
   - 色飽和度較低
2. Blue Mini-LED + QD Film
   - 藍光 LED
   - 色域大
   - 響應快
   - 色彩純度高
3. RGB Mini-LED
   - 直接三色 LED

- 高色飽和
- 高對比
- 成本高
- 三色老化速率不同
- 校正難度高

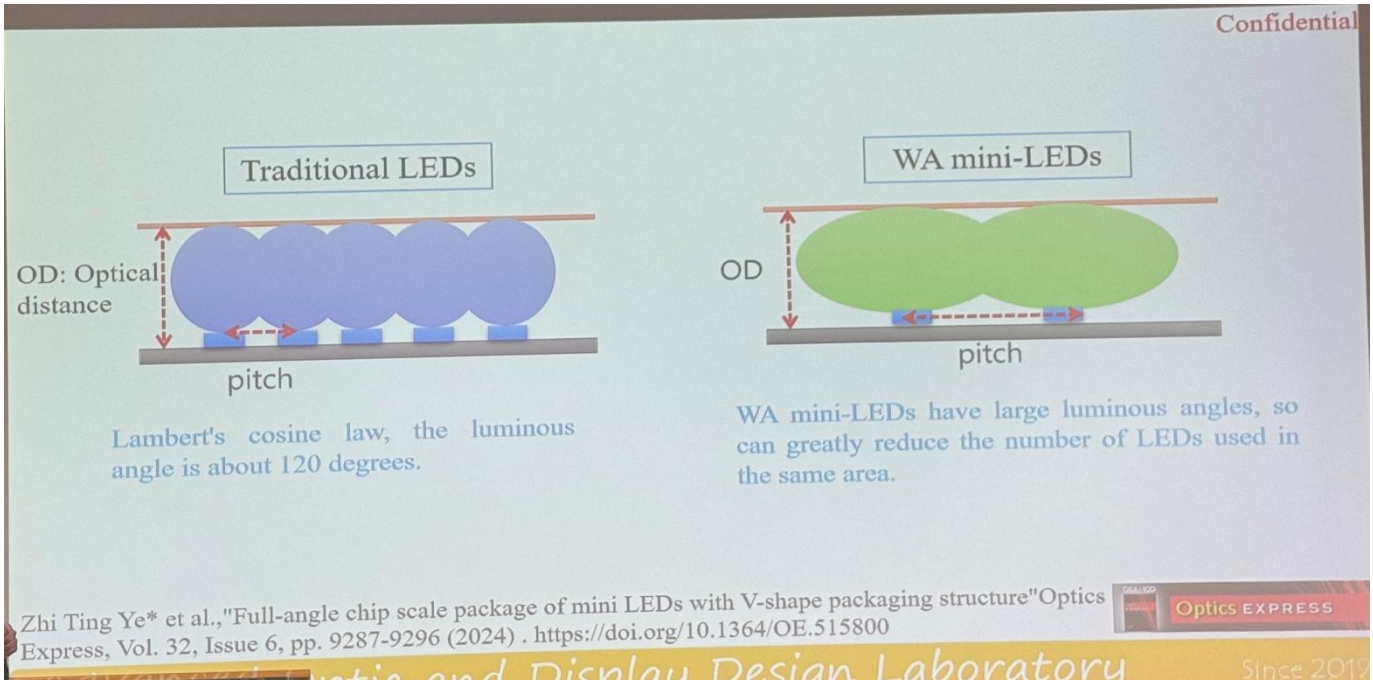


ZOD（Zero Optical Distance）

- 微透鏡陣列，LED 光場快速展開，幾乎不用光學距離
- 模組薄

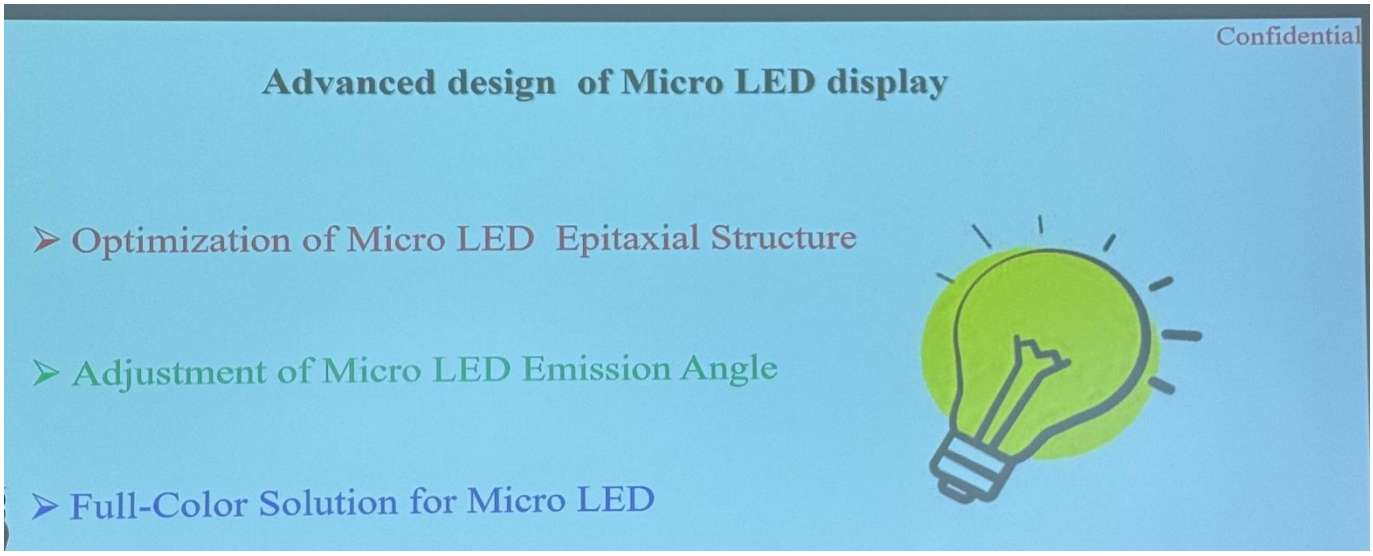


- 傳統邊緣光 BLU 亮區周圍會漏光，形成 halo（光暈）
- Mini-LED + LGMS：光分布變窄，邊緣斜率變陡，黑區更黑



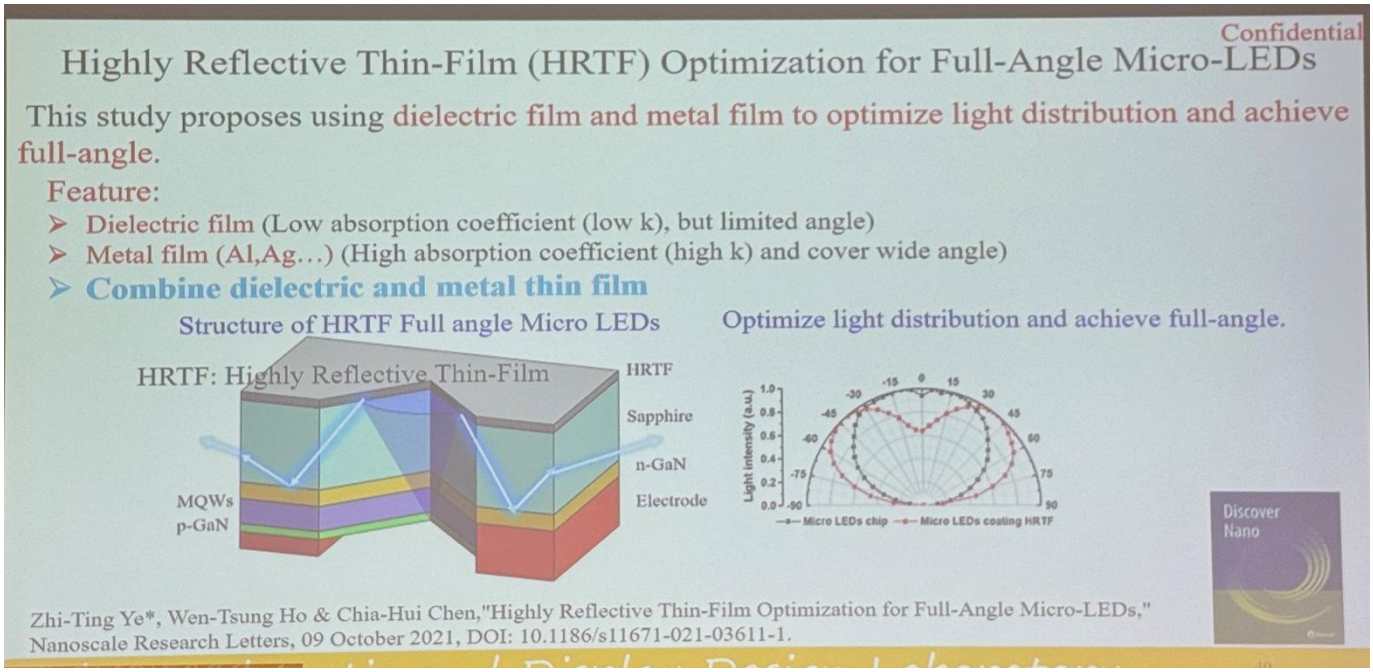
- 傳統 LED
  - Lambert cosine
  - 發光角約 120°
- WA mini-LED
  - 發光角變大
  - 光場更展開

1. 減少 LED 顆數
2. 降低成本
3. 保持均勻度



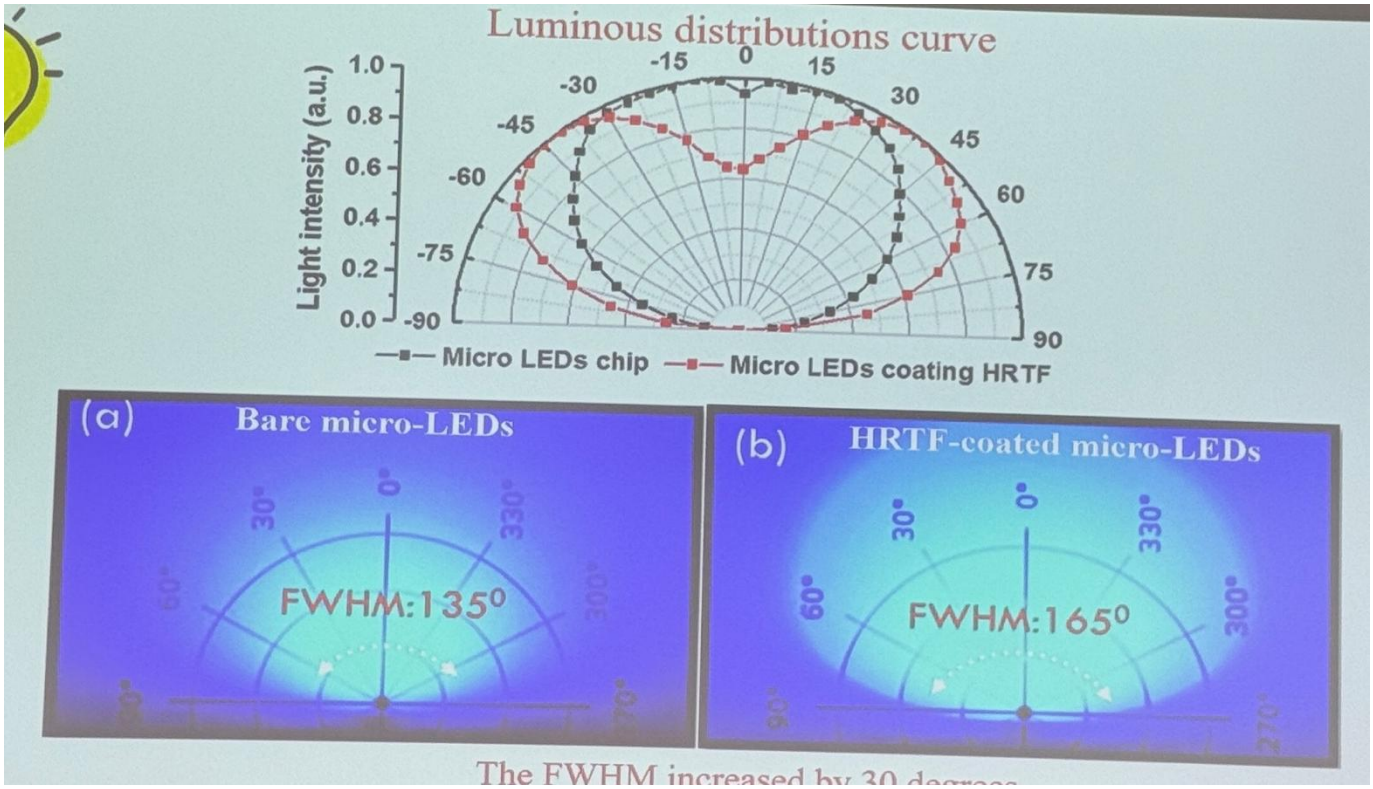
Micro LED 進階設計

- 最佳化外延結構
- 調整發光角度
- 全彩解決方案



HRTF（Highly Reflective Thin Film）

- 改善出光角度
- 提高全角發光
- 提升光提取效率



- Bare micro-LED：FWHM = 135°
- HRTF-coated：FWHM = 165°
- 增加 30°